

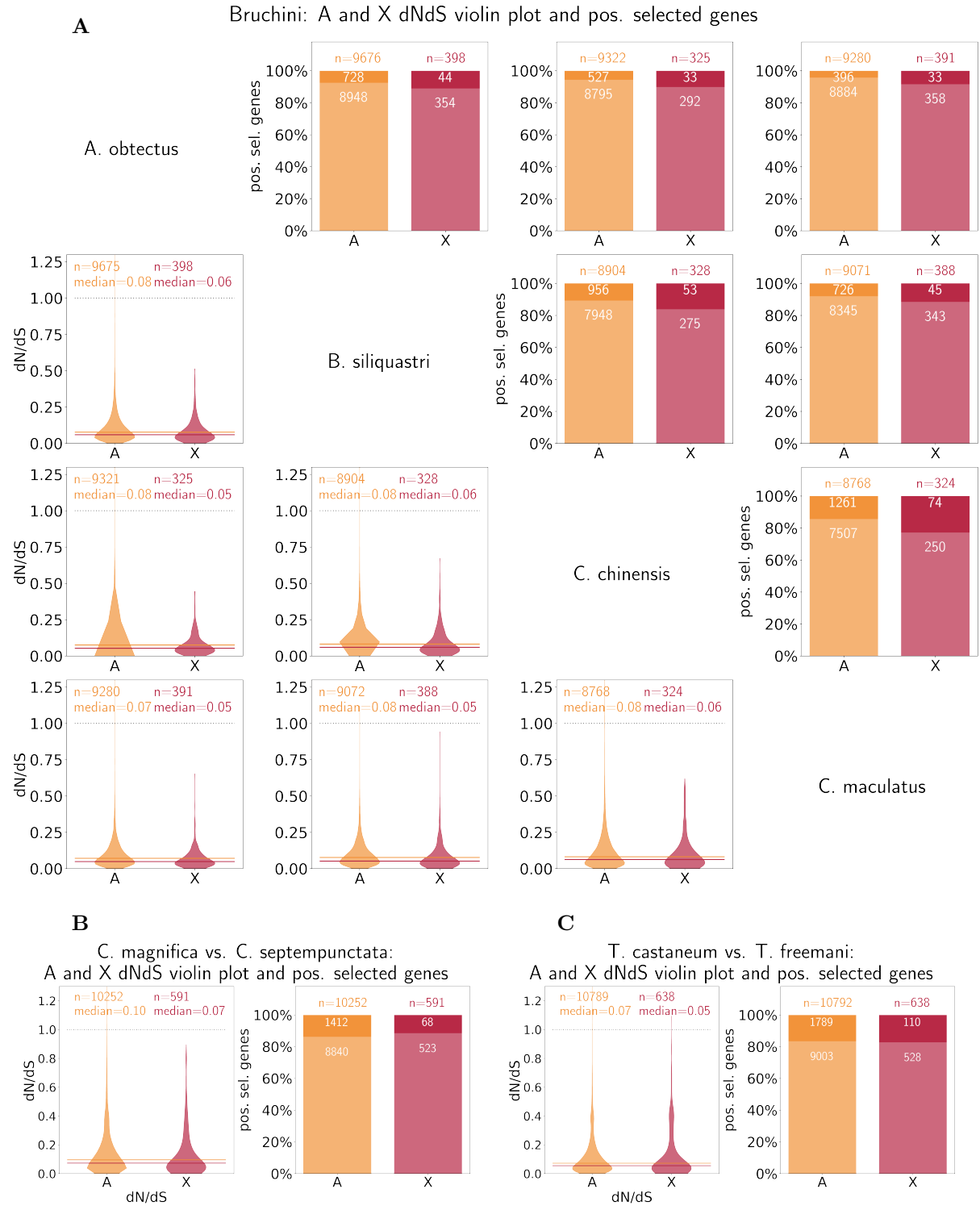


Figure 1: d_N/d_S and abundance of positively selected genes. *Violin plots:* Pairwise comparisons between members of three species groups with violin plots of d_N/d_S values for X-linked and autosomal 1-to-1 orthologs, and scatterplots of d_S vs. d_N . Permutation tests ($n=10000$) show that d_N/d_S is significantly lower for X-linked orthologs in all comparisons within *Bruchini* (A) and *Coccinella* (B), but not *Tribolium* (C) (see Fig. S7 for permutation test results). *Bar plots:* The bar plot show the percentage of genes that have codons under positive selection (darker section on top) according to paml site-model analysis (likelihood ratio test comparison between paml site models M1a and M2a). Permutation tests ($n=10000$) show that *Bruchini* (A) have an enrichment of positively selected X-linked orthologs in all pairwise comparisons, while *Coccinella* (B) and *Tribolium* (C) show no significant difference (see Fig. S8).

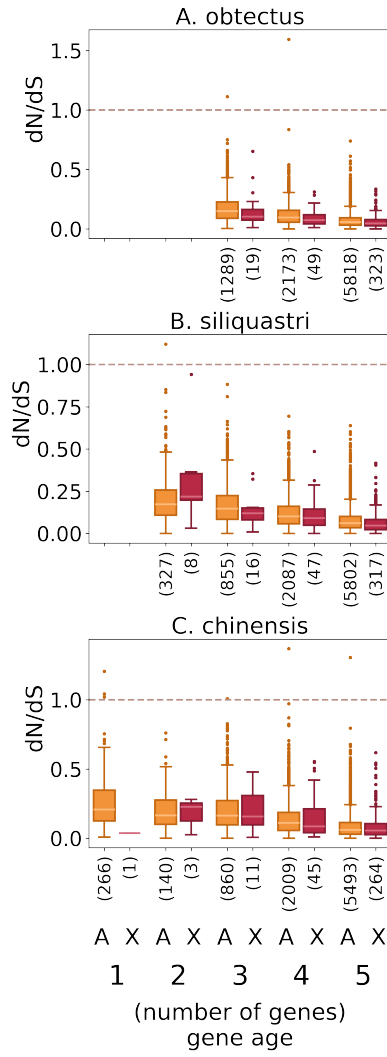


Figure 2: **Gene age:** Pairwise comparisons of *C. maculatus* with *A. obtectus*, *B. siliquastri* and *C. chinensis* respectively. X-axes ticks show sample size in parentheses, as well as gene age (less conserved orthologs of age 1 to most conserved orthologs of age 5), and autosomal or X-linkage of orthologs (A or X). The box plots show d_N/d_S for autosomal (yellow) or X-linked (red) orthologs by gene age for *C. maculatus* compared to three other textitBruchini species.

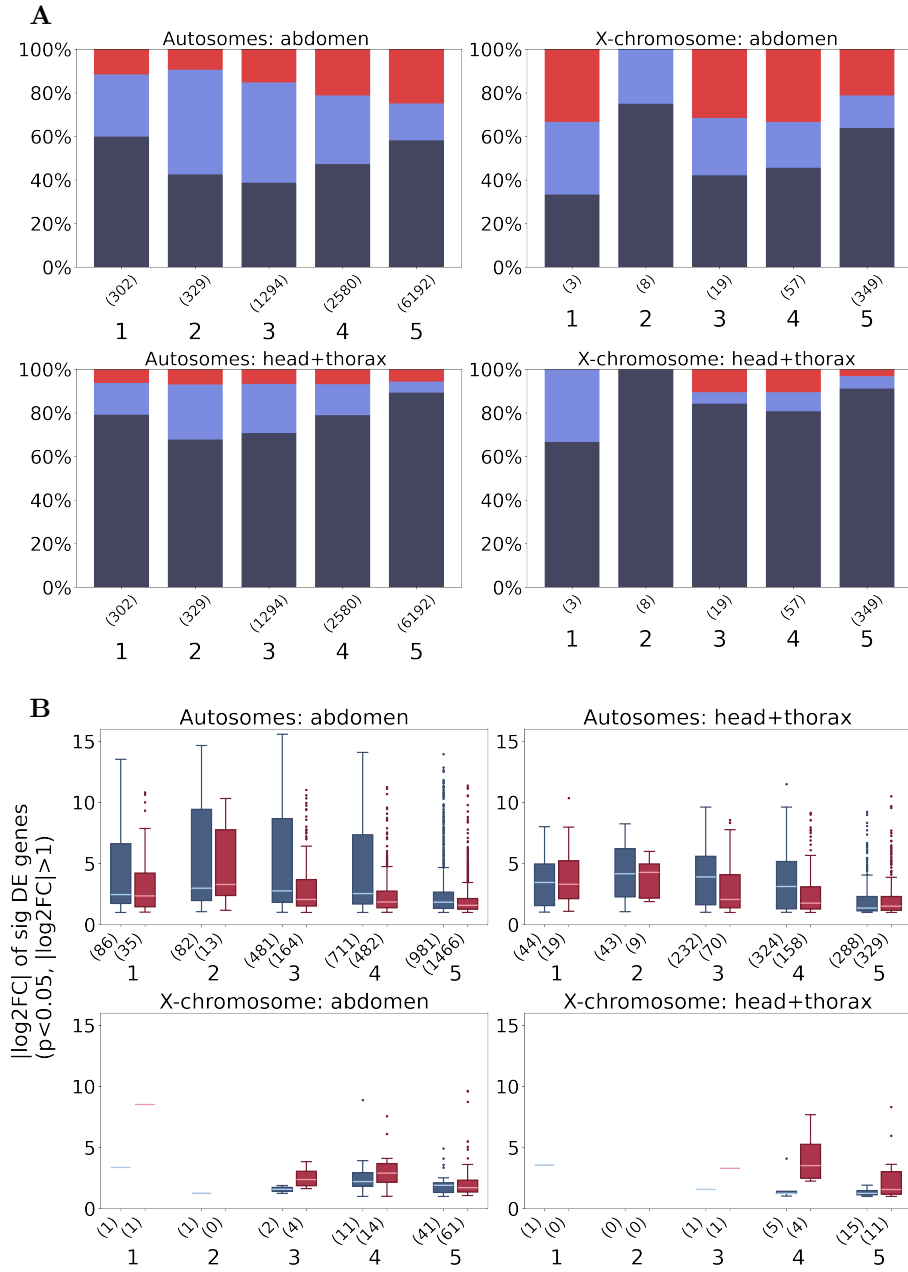


Figure 3: **gene age and sex-bias analysis**: X-axes ticks show sample size in parentheses, as well as gene age (1 to 5), and autosomal or X-linkage of orthologs (A or X). **(A)**: proportion of sex-bias categories for each gene age category according to abdominal or head and thorax tissues, as well as autosomal or X-linked orthologs. Unbiased in grey, female-biased in red and male-biased in blue. **(B)**: Absolute log₂FC for significantly male-biased (blue) or female-biased (red) orthologs of *C. maculatus* and *C. chinensis* for increasing gene age. Split by autosomal and X-linked orthologs, as well as log₂FC determined from abdominal tissue (reproductive) or head and thorax tissue (somatic).

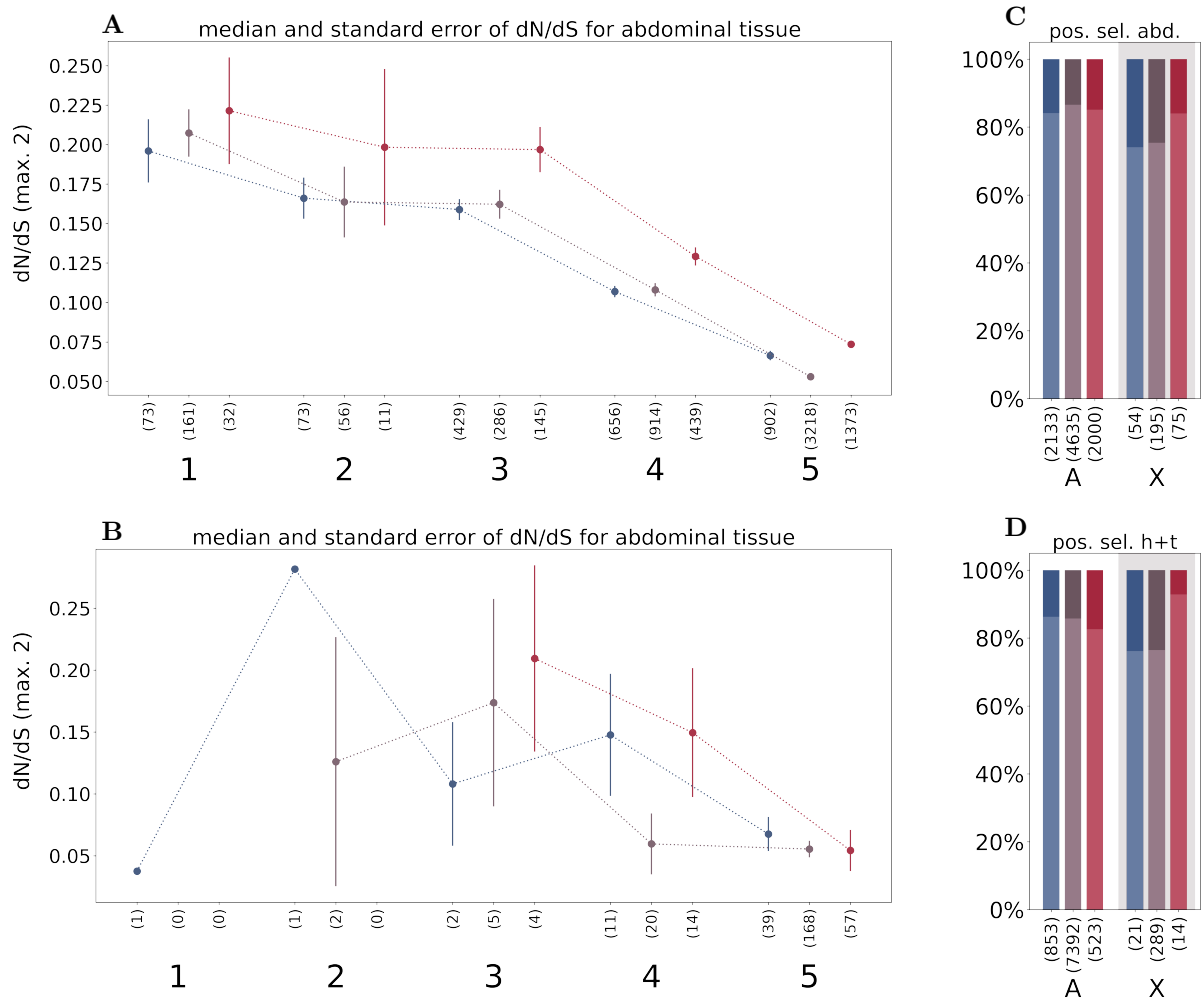


Figure 4: **Impact of gene age and sex-bias on d_N/d_S and positively selected genes:** X-axes ticks show sample size in parentheses, as well as gene age from less conserved orthologs (age 1) to highly conserved orthologs (age 5), or autosomal or X-linkage of orthologs (A or X). (A,B): median d_N/d_S and standard error of autosomal orthologs, split by gene age as well as sex-bias category according to abdominal tissues with male-bias being blue, unbiased being grey and female-biased being red. (A): autosomal, (B): X-linked. (C,D): Proportion of genes under positive selection according to paml site model analysis (darker section) split by Autosomal and X-linked orthologs, as well as male-biased (blue), unbiased (grey) and female-biased (red) orthologs according to either abdominal tissue (reproductive, C) or head and thorax tissue (somatic, D).